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HCQNSC: a hybrid classical-quantum neural network for text sentiment classification

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Abstract

Combining variational quantum circuits with classical neural network models is a potential direction for quantum natural language processing (QNLP) in the NISQ (Noisy Intermediate-Scale Quantum) era. We propose a hybrid classical-quantum neural network that enhances a transformer architecture in two key ways: first, by incorporating optimized variational quantum circuits into the self-attention and multilayer perceptron components, and second, by integrating a hybrid convolutional module for effective dimensionality reduction which can reduce the quantum resources required for subsequent layers and enhances the model's trainability by mitigating issues like the barren plateau phenomenon, thus making it feasible to construct a deeper and more powerful network architecture. Our experiments on three widely-used sentiment analysis datasets demonstrate that the proposed model achieves comparable or even superior performance in terms of accuracy and F1 score compared to classical transformer-based sentiment classification models with similar structures, while reducing the number of parameters through the utilization of quantum advantages.

Keywords: Quantum Sentiment Classification; Variational Quantum Circuits; Quantum Neural Network; Quantum Machine Learning

1 Introduction

Quantum computing [1] exploits the principles of quantum mechanics [2] to process information in ways that differ from classical systems. Through quantum properties such as superposition, entanglement, and interference, it has demonstrated potential for accelerating computation in certain problem classes [3]. The field has garnered considerable interest in both academic and industrial settings, catalyzing advancements across various disciplines, including cryptography [4], chemistry [5], and machine learning [6–8]. As hardware capabilities continue to evolve, there is a growing emphasis on designing quantum algorithms that can outperform classical counterparts in areas like optimization [9], data analysis, and complex pattern recognition [10], driving forward the development of powerful, application-specific solutions in fields that demand high computational resources.

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Quantum natural language processing (QNLP) [11, 12] is a growing research area that combines quantum computing with artificial intelligence[13] for language-related tasks. As quantum computers become more available, researchers have begun applying its computational properties to NLP problems. While traditional NLP has relied on classical neural-network-based models to interpret and analyze language, QNLP aims to exploit the principles of quantum mechanics, such as superposition and entanglement, to represent linguistic structures more effectively and potentially surpass classical approaches in terms of expressivity and efficiency.

Despite this potential, the current limitations of quantum hardware, often categorized under the Noisy Intermediate-Scale Quantum (NISQ) era [14], restrict the scale and depth of purely quantum approaches. NISQ devices are prone to errors and limited in qubit count, making it challenging to perform large-scale NLP tasks entirely on quantum systems. Hybrid models that combine classical and quantum components therefore offer a practical route to exploit quantum advantages within present hardware constraints [15, 16]. In such models, selected classical components are replaced by quantum subroutines, keeping the overall computation tractable.

In this paper, we introduce a Hybrid Classical-Quantum Neural Network for Sentiment Classification called HCQNSC. The core of our approach involves enhancing the classical transformer-based text classification architecture by substituting its classical linear transformations with variational quantum circuits (VQCs). Specifically, the high-parameter matrix multiplications within both the self-attention and multilayer perceptron components are replaced by more parameter-efficient VQCs. To address the challenge of scaling this approach to deeper networks, we further integrate a hybrid convolutional module into the architecture's later layers. This module performs progressive dimensionality reduction, a crucial step that lowers the quantum resource requirements for subsequent layers. By enhancing the overall model's trainability and mitigating issues like the barren plateau phenomenon, this design makes it feasible to construct a deeper network capable of learning more complex features for the classification task.

We conducted comprehensive experiments on three commonly used sentiment classification datasets (SST-2 [17], IMDB [18], Yelp). The results show that HCQNSC consistently outperforms classical non-transformer-based models and other existing hybrid quantum NLP models, demonstrating the effectiveness of our architecture. When compared against its structurally equivalent classical counterpart (CTNSC), our quantum-enhanced approach could deliver a highly competitive and statistically comparable performance while using significantly fewer parameters. By leveraging quantum principles, the model offers a dramatic reduction in parameter complexity without a statistically significant sacrifice in performance, showcasing its potential as a viable and resource-efficient solution for NLP tasks in the NISQ era.

2 Related work

Quantum-inspired approaches have gained attention in Natural Language Processing (NLP) due to their potential to model linguistic nuances through principles derived from quantum mechanics. Early quantum-inspired models, such as Sordoni et al. [19] and Li et al. [20], applied quantum probabilistic frameworks to NLP tasks, utilizing density matrices to capture complex text relationships within a classical computing setup. Subsequent developments like the Neural Network Based Quantum-Like Language Model (NNQLM) by

Zhang et al. [21] employed density matrices to represent sentence similarity within neural network architectures. Other approaches, like Li et al.'s complex embedding mixture network [22], leveraged quantum interference principles to derive nuanced phrase meanings, showing promising performance improvements in sentence classification tasks. More recently, the ICWE-QNN and CICWE-QNN models proposed by Shi et al. [23], demonstrated the feasibility of integrating complex-valued embeddings and deep learning techniques to address linguistic feature loss, achieving superior performance in binary text classification tasks on benchmark datasets. However, while these models are inspired by quantum mechanics, they remain entirely classical, utilizing quantum analogies within traditional neural architectures rather than leveraging genuine quantum computation.

On the frontier of actual quantum computation for NLP, quantum-native models attempt to employ quantum hardware directly for language processing tasks. However, these models face inherent limitations due to the constraints of Noisy Intermediate-Scale Quantum (NISQ) devices. Meichanetzidis et al. [24] laid foundational work in this area by introducing a full-stack pipeline that leverages DisCoCat (Distributional Compositional Categorical) semantics to model grammatical structures as quantum circuits, thus allowing for the encoding of syntactic and semantic structures within a quantum computational framework. This model provided a means of mapping linguistic representations onto quantum circuits compatible with near-term quantum devices. Lorenz et al. [25] expanded on this by implementing the first practical NLP tasks on NISQ devices for datasets exceeding 100 sentences, demonstrating the applicability of quantum-native approaches in limited, syntax-sensitive tasks. Nevertheless, the small-scale application scope and high error rates on current hardware present significant barriers to scalability in complex NLP tasks, confining quantum-native NLP primarily to experimental environments with simple datasets and reduced problem complexity.

Given the current limitations of NISQ devices, classical-quantum hybrid approaches have emerged as the most viable strategy for implementing quantum advantages in NLP. These models typically involve replacing specific components of classical architectures with quantum circuits, thereby leveraging quantum properties without overreliance on full quantum processing. Henderson et al. [26] introduced Quanvolutional Neural Networks, which incorporate quantum convolutional layers into classical Convolutional Neural Networks (CNNs), demonstrating enhanced feature extraction capabilities with modest quantum requirements, thus aligning with the current state of quantum technology. In a similar vein, Yang et al. [27] and Di Sipio et al. [28] proposed quantum-enhanced frameworks for automatic speech recognition and part-of-speech tagging, where quantum circuits are used for feature extraction or processing, with the classical model handling the remainder of the task. This hybrid design mitigates the limitations of quantum hardware by restricting quantum computation to specific sub-tasks. The most recent work by Yang et al. [29] integrated quantum temporal convolution layers within a BERT model to enhance text classification, showcasing the practical potential of hybrid models by achieving competitive performance on various classification datasets. Li et al. [30] extended this concept with Quantum Self-Attention Neural Networks (QSANN), which implement quantum self-attention layers, demonstrating robustness to quantum noise and scalability on larger datasets.

3 Preliminaries

3.1 Classical transformer

The classical Transformer model, introduced by Vaswani et al. in 2017 [31], has become a foundational architecture in deep learning, particularly for natural language processing (NLP) tasks. Its distinctive mechanism—self-attention—enables the model to weigh the importance of different words in a sentence relative to each other, allowing it to capture complex dependencies in the data, even over long distances.

A Transformer layer primarily consists of two components: Multi-Head Self-Attention Mechanism and Feed-Forward Neural Network (FFN). Each of these components is wrapped with residual connections and layer normalization, facilitating efficient training and stabilizing gradients. The self-attention mechanism is at the heart of the Transformer. In NLP, given a sequence of input words, the model generates three matrices for each word embedding: a query Q, a key K, and a value V. The attention score for each word in relation to another word is computed by taking the dot product between their query and key vectors. Mathematically, the attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

where d_k is the dimension of the key vectors, acting as a scaling factor to prevent excessively large values in the dot product. The resulting scores allow the model to selectively focus on specific parts of the input sequence, essential for tasks requiring nuanced contextual understanding. In multi-head attention, the model projects the queries, keys, and values h times with different learned linear transformations. Each head processes the input sequence in parallel, capturing distinct aspects of the data. The outputs of all heads are then concatenated and transformed once more, yielding a rich representation of the input:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O \tag{2}$$

where each $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$ with learnable weight matrices W_i^Q, W_i^K, W_i^V and W^O. After the multi-head attention layer, the output is fed into a position-wise feed-forward neural network, which consists of two linear transformations with a ReLU or GELU activation in between:

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2 \tag{3}$$

This component allows the Transformer to apply non-linearity and enhance its expressive power by transforming the attended information. The classical Transformer model, particularly its encoder, processes the input sequence by stacking multiple layers of self-attention and FFN blocks, enabling it to learn increasingly abstract representations. Each layer refines the understanding of the sequence, with the final layer output capturing the context-rich information needed for downstream tasks such as sentiment classification, translation, and more.

3.2 Variational quantum circuits

VQCs [32] form the backbone of many hybrid quantum-classical models, especially in the context of the Noisy Intermediate-Scale Quantum (NISQ) era. These circuits utilize parameterized quantum gates to create a tunable quantum state, which can be optimized by adjusting the parameters during training, which also become the foundation of quantum neural networks [33, 34]. A VQC is typically composed of multiple layers of single-qubit and two-qubit gates, and each gate can be parameterized to allow fine-tuning of the circuit during the optimization process. Single-qubit gates operate on individual qubits, modifying their states based on specific rotations. Parameterized single-qubit gates play an essential role in creating the desired quantum state, as they allow controlled transformations that are adaptable through parameter tuning. The $R_X(\theta)$, $R_Y(\theta)$, and $R_Z(\theta)$ gates are the parameterized single-qubit gate which perform rotations around the X-axis, Y-axis, and Z-axis of the Bloch sphere by an angle θ, respectively.

$$
\begin{aligned}
R_X(\theta) &= \begin{bmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{bmatrix} \\
R_Y(\theta) &= \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix} \\
R_Z(\theta) &= \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}
\end{aligned}
\tag{4}
$$

Two-qubit gates are essential for creating entanglement between qubits, a unique quantum property where the state of one qubit becomes intrinsically linked to another. Entanglement is critical for the expressiveness of VQCs, as it allows the circuit to capture complex, correlated structures in data. The CNOT gate flips the state of the target qubit if the control qubit is in the state $|1\rangle$. This gate is frequently used to create entangled states and serves as a fundamental building block in constructing entangled circuits. Controlled rotation gates such as $CR_X(\phi)$, $CR_Y(\phi)$, and $CR_Z(\phi)$ apply a parameterized rotation to the target qubit, conditioned on the control qubit being in the $|1\rangle$ state.

$$
\begin{aligned}
CR_X(\phi) &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos(\phi/2) & -i\sin(\phi/2) \\ 0 & 0 & -i\sin(\phi/2) & \cos(\phi/2) \end{bmatrix}. \\
CR_Y(\phi) &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos(\phi/2) & -\sin(\phi/2) \\ 0 & 0 & \sin(\phi/2) & \cos(\phi/2) \end{bmatrix}. \\
CR_Z(\phi) &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & e^{-i\phi/2} & 0 \\ 0 & 0 & 0 & e^{i\phi/2} \end{bmatrix}.
\end{aligned}
\tag{5}
$$

According to the research of Sim et al. [35], the effectiveness of a VQC depends heavily on two critical characteristics: Expressibility and Entangling Capability. Expressibility is

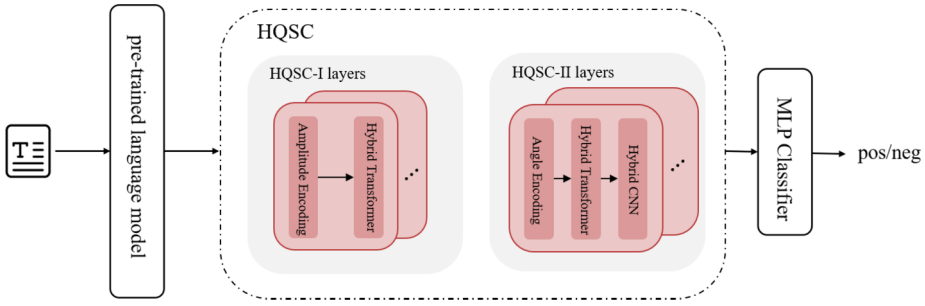


Figure 1 Model Structure

the ability of the circuit to represent a wide variety of quantum states. Higher expressibility enables the circuit to approximate a broader range of functions, which is beneficial for capturing complex patterns in data. Entangling Capability measures the extent to which the circuit can produce entangled states. High entangling capability allows the circuit to model intricate correlations within data, which is essential for tasks that rely on capturing multi-variable dependencies, like sentiment analysis. Both expressibility and entangling capability can be enhanced by choosing appropriate parameterized single-qubit and two-qubit gates, as well as by configuring them in multi-layer structures. These properties are pivotal for ensuring that the VQC can sufficiently approximate the target function in hybrid quantum-classical algorithms.

By incorporating these gates and optimizing the parameters through a hybrid training approach, VQCs can provide a flexible yet powerful framework for quantum-enhanced NLP models, such as the sentiment classification model discussed in this paper.

4 Proposed model

The Hybrid Neural Network proposed for sentiment classification in this work (HCQNSC) integrates quantum and classical neural network components, creating a unique hybrid quantum-classical framework (Fig. 1). The HCQNSC processes token embeddings generated by a classical pretrained model, with its architecture structured around two types of Hybrid Quantum Sentiment Classifier (HQSC) layers: HQSC-I and HQSC-II. Each type utilizes the model's core modules—the quantum encoding module, the hybrid transformer module, and the hybrid convolutional module—in different ways tailored to the dimensionality of the embeddings.

HQSC-I Layer: This layer is designed for processing high-dimensional classical word embeddings. The Quantum Encoding Module in the HQSC-I layer uses Amplitude Encoding to transform these high-dimensional feature vectors into quantum states. In the Hybrid Transformer Module, simpler VQCs with a relatively low number of parameters are employed to capture contextual relationships within the embeddings. This layer does not include the Hybrid Convolutional Module; instead, dimensionality reduction is achieved through measurement after processing, significantly lowering the dimensionality of the word embeddings before they are passed to the next stage.

HQSC-II Layer: The HQSC-II layer is designed for embeddings that have undergone substantial dimensionality reduction in the HQSC-I layers. In this layer, the Quantum Encoding Module uses Angle Encoding, suitable for the lower-dimensional embeddings. The Hybrid Transformer Module incorporates more complex VQCs with a larger number of

parameters, leveraging the increased representational capacity to handle refined contextual transformations. Additionally, this layer includes the Hybrid Convolutional Module, which performs gradual, layer-by-layer dimensionality reduction on the feature vector. This controlled compression further optimizes the efficiency of the quantum transformer while retaining essential features for sentiment analysis.

As the token embeddings traverse through the sequence of HQSC layers, the dimensionality is progressively reduced. Starting with the high-dimensional embeddings processed by HQSC-I layers, the model encodes these into quantum states, applies quantum transformations, and reduces dimensionality through measurement. These embeddings then enter the HQSC-II layers, where more sophisticated quantum circuits and convolutional processing achieve refined transformations on the reduced-dimensionality data. This layered architecture enables efficient and effective sentiment classification by combining quantum and classical methods, leveraging quantum resources under the constraints of Noisy Intermediate-Scale Quantum (NISQ) hardware.

4.1 Classical pre-trained language model for word embedding

In our Hybrid Model (HCQNSC), we utilize DistilBERT [36] as a pre-trained language model to generate contextual word embeddings from the input text sequences. DistilBERT, a smaller and faster variant of the BERT model, has been distilled to retain much of the linguistic and contextual knowledge of BERT while being computationally efficient. This makes it particularly suitable for hybrid quantum-classical models where resource constraints are crucial.

The word embedding process in DistilBERT follows a multi-step architecture inspired by Transformer models. Given an input sequence of tokens $x = (x_1, \ldots, x_T)$, each token is initially transformed into a high-dimensional embedding vector through three components: WordPiece embeddings, positional embeddings, and segment embeddings. The resulting vectors capture both the semantic meaning of each word and their positions within the sentence, enabling the model to understand context and relationships between words.

Mathematically, the embedding process in DistilBERT can be represented as:

$$H = \text{DistilBERT}(x_1, \ldots, x_T), \tag{6}$$

where $H = (h_1, \ldots, h_T)$ and each h_i is a contextual semantic embedding for the token x_i. Here, h_i encompasses not only the word's intrinsic meaning but also its syntactic and semantic relationships with other words in the sentence. These embeddings serve as the input to the subsequent quantum modules in the HCQNSC, where quantum processing layers further refine and transform the data for sentiment classification tasks.

By leveraging DistilBERT's pre-trained knowledge, our model can efficiently encode complex linguistic structures into dense representations, providing a robust foundation for the hybrid quantum-classical processing that follows. This approach optimally balances the high-dimensional embedding capabilities of classical language models with the expressive power of quantum circuits in later stages of the HCQNSC.

4.2 Quantum encoding module

The Quantum Encoding Module serves as the bridge between classical embeddings generated by DistilBERT and the quantum processing layers within our hybrid model. By encod-

ing high-dimensional classical data into quantum states, we leverage the quantum feature space to efficiently represent and process linguistic information. This section details the encoding strategies employed: Amplitude Encoding and Angle Encoding.

4.2.1 Amplitude encoding

Amplitude Encoding is an efficient technique for mapping classical data into quantum states by utilizing the amplitudes of quantum vectors. Given an n-dimensional normalized classical vector $x = (x_1, x_2, \ldots, x_n)$, where each component x_i satisfies $\sum_{i=1}^{n} |x_i|^2 = 1$, the data is encoded into the quantum state amplitudes, allowing for compact data representation within an exponentially large Hilbert space. Specifically, an n-dimensional vector is encoded using at least $\log_2(n)$ qubits, capturing all elements of the vector within a single quantum state:

$$|x\rangle = \sum_{i=1}^{n} x_i |i\rangle \tag{7}$$

This encoding method is particularly advantageous in quantum computation as it allows the model to encode large datasets into quantum amplitudes efficiently. The ability to represent the full vector within a few qubits optimizes memory use and computational resources, thus enhancing processing capability within the quantum layers of our model.

4.2.2 Angle encoding

Angle Encoding, on the other hand, maps data onto the rotational angles of quantum gates, providing a straightforward way to encode real values from classical data. For a classical vector $x = (x_1, x_2, \ldots, x_n)$, each element x_i is used as a parameter for a rotation applied to the i-th qubit. In our model, we employ R_y rotations, where each data point x_i defines the rotation angle for the corresponding qubit:

$$|x\rangle = R_Y(x_1) \otimes R_Y(x_2) \otimes \cdots \otimes R_Y(x_n) |0\rangle^{\otimes n} \tag{8}$$

This encoding approach is particularly suitable for embeddings that have undergone dimensionality reduction, as it allows each feature to directly control the rotation of its associated qubit, providing an interpretable and efficient way to encode the reduced dimensions. Furthermore, the simplicity of angle encoding facilitates the integration of these encoded states within our quantum variational circuits, contributing to the flexibility and adaptability of the Quantum Encoding Module in processing sentiment-related information. Both Amplitude Encoding and Angle Encoding play crucial roles within the Hybrid Quantum Language Model, enabling it to fully exploit quantum states to capture and process complex linguistic structures for sentiment classification tasks.

4.3 The hybrid transformer module

In the Hybrid Transformer Module of the HCQNSC model, the key principle is to replace the classical parameter matrices W^Q, W^K, W^V, W^O in the self-attention mechanism and the parameter matrices in the feed-forward network (FFN) layer with VQCs as Fig. 2 shows. In the classical transformer, the function of these parameter matrices is to act as learnable linear transformations. Specifically, within the self-attention mechanism,

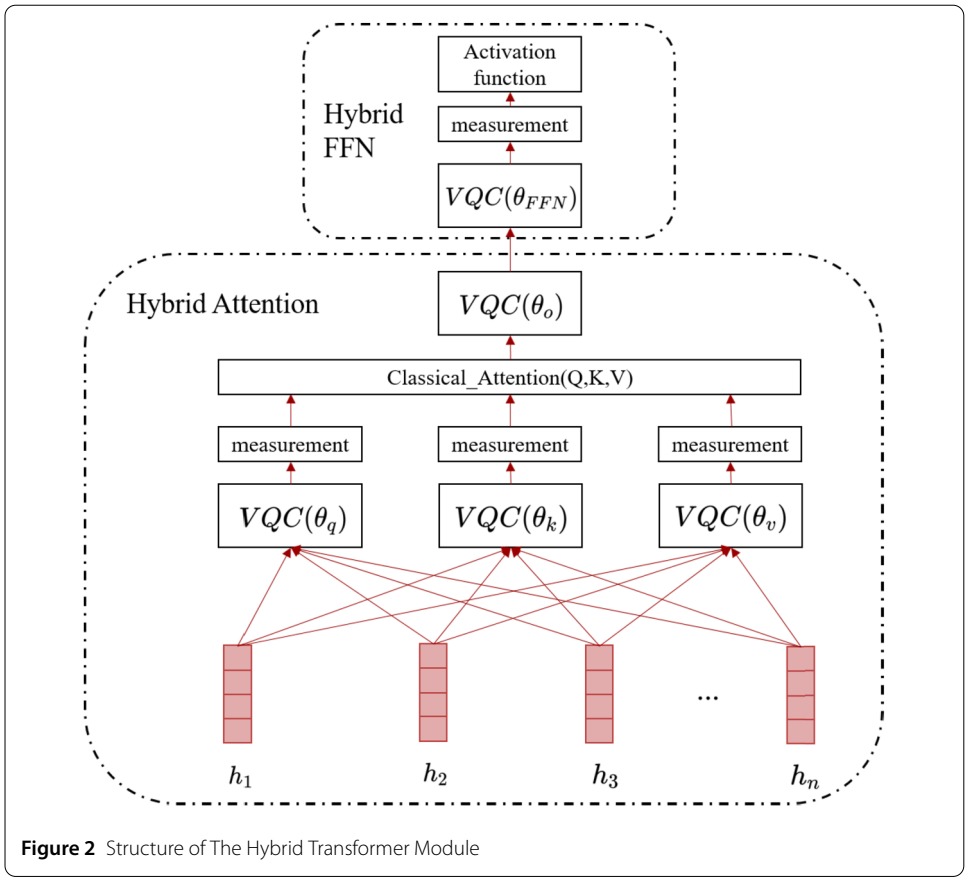
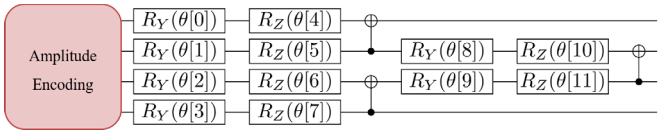


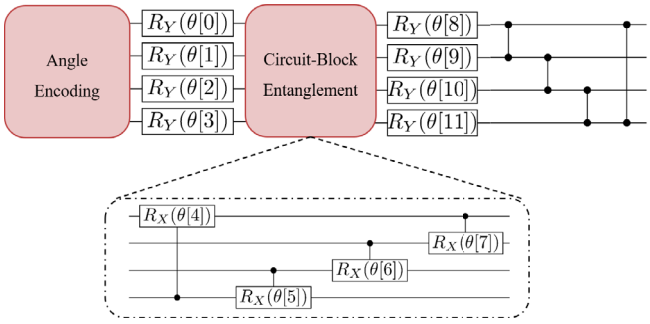
Figure 2 Structure of The Hybrid Transformer Module

they project the input embedding vectors into different semantic subspaces—namely, the Query, Key, and Value spaces—which are essential for the subsequent attention score calculation. In the FFN, they provide further linear transformations between the non-linear activation functions. This quantum substitution offers a significant reduction in the model's parameter count compared to classical transformers. The rationale behind this approach is that while classical parameter matrices require $O(n^2)$ parameters to process n-dimensional vectors, quantum circuits can represent transformations of comparable representational power using only $O(n)$ parameters. This is achieved by leveraging the exponential computational capacity of the quantum Hilbert space, which allows the model to encode the required transformations in a more compact and efficient manner. The design of VQCs varies between HQSC-I and HQSC-II layers to accommodate different processing requirements.

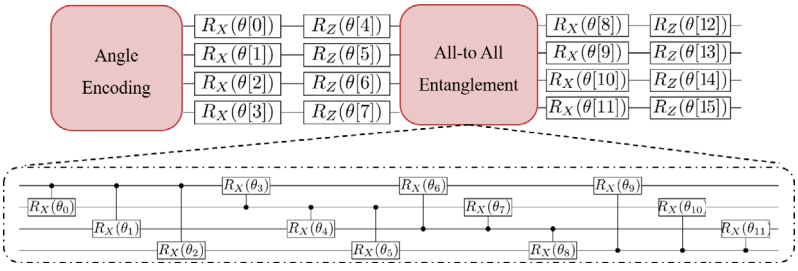
The HQSC-I layer processes high-dimensional input embeddings, necessitating quantum circuits with high qubit counts. To maintain efficiency on NISQ devices, we design these circuits with emphasis on simplicity and parameter efficiency while maintaining strong feature encoding capabilities. Dimensionality reduction in HQSC-I is achieved through controlled measurement of quantum states, allowing flexible compression when converting quantum states back to classical feature vectors. This is done by selectively measuring only a subset of the total quantum bits, thereby reducing the dimensions of the classical feature vectors while retaining the key information encoded in the quantum states.



(a) VQC for Linear Transformation in HQSC-I Self-Attention (Nearest-neighbor connectivity)



(b) VQC for Linear Transformation in HQSC-II Self-Attention with Moderate Qubit Count (Circuit-block connectivity)



(c) VQC for Linear Transformation in HQSC-II Self-Attention with Low Qubit Count (All-to-all connectivity)

Figure 3 Quantum Circuits Used in Hybrid Transformer Module

HQSC-II layer processes reduced-dimensional features from HQSC-I, enabling more sophisticated quantum circuit architectures. The attention layer implements two distinct VQC structures based on a qubit number threshold ϵ.

The VQCs used in this model are as follows:

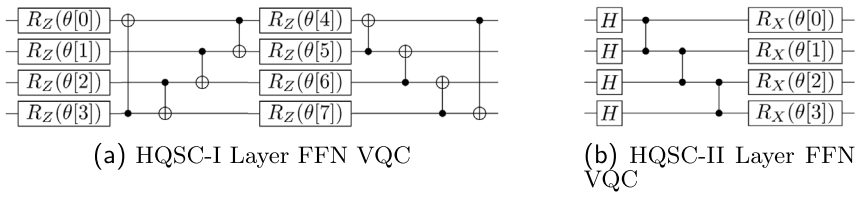
4.3.1 VQCs for linear transformations in self-attention

i Circuit of HQSC-I Layer (Fig. 3(a)):

- Implements a Nearest-neighbor connectivity pattern.
- Uses non-parameterized CNOT gates for two-qubit operations.
- Incorporates parameterized R_Y and R_Z rotations in a layered structure.
- Achieves strong entangling capability with moderate expressibility.
- Minimizes parameter count through simple gate selection.

ii Circuits of HQSC-II Layer:

- For qubit counts above ϵ (Fig. 3(b)):
  - Employs parameterized CR_X gates.
  - Uses Circuit-block connectivity pattern.
  - Maintains moderate circuit depth and parameter count.



(a) HQSC-I Layer FFN VQC (b) HQSC-II Layer FFN VQC

Figure 4 Quantum Circuits Used in FFN Module

– Provides balanced expressibility and entangling capability.

- For qubit counts below ϵ (Fig. 3(c)):
  – Implements parameterized CR_X gates.
  – Features All-to-all connectivity pattern.
  – Utilizes deeper circuit structure.
  – Maximizes both expressibility and entangling capability.

4.3.2 FFN module VQCs

i Circuit of HQSC-I Layer (Fig. 4(a)):
- Employs Hadamard gates for initial state preparation.
- Uses Controlled-Z gates in a Nearest-neighbor connectivity pattern.
- Achieves high entangling capability despite minimal circuit depth.
- Maintains simplicity as feature encoding primarily occurs in attention layer.

ii Circuit of HQSC-II Layer (Fig. 4(b)):
- employs a simplified quantum circuit structure.
- optimized for shallow depth while supporting efficient computation across multiple transformer layers.
- particularly suitable for processing the reduced-dimensional features.

This hybrid quantum-classical architecture effectively balances the computational advantages of quantum circuits with NISQ device constraints, enabling efficient processing of linguistic features for sentiment classification tasks.

4.4 Hybrid convolutional module

The Hybrid Convolutional Module which aims to leverage the advantages of quantum computation and classical convolutional neural networks by replacing the classical convolution kernels with customized quantum circuits is incorporated into the HQSC-II layer of the proposed HCQNSC. After the hybrid transformer layers in the HQSC-II stage have processed the classical feature vectors obtained through the mean-measurement of quantum states, the Hybrid Convolutional Module further refines and compresses the features. The key idea is to substitute the classical convolution operation with a quantum circuit-based equivalent, allowing the model to benefit from the expressive power and efficiency of quantum processing.

The architecture of the Hybrid Convolutional Module is shown in Fig. 5, we use a 4-qubit VQC to do the quantum convolution operation, the selected quantum circuit for quantum convolution operation exhibits strong entangling capability while maintaining a relatively low parameter count in the single-layer case. By measuring the Pauli-Z expectation values of a specific qubit within this quantum circuit, the model can effectively implement

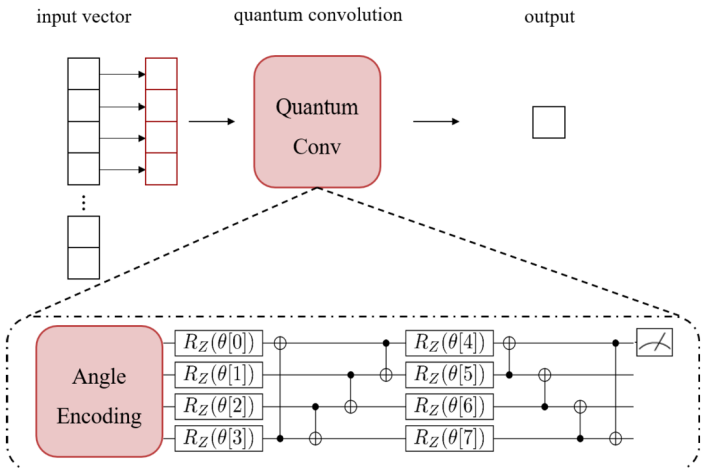


Figure 5 Structure of Quantum Convolutional Module

the quantum convolution operation and further compress the feature representations. The integration of this Hybrid Convolutional Module into the HQSC-II layer enables the model to progressively reduce the dimensionality of the linguistic features while preserving the essential information for sentiment analysis. This hybrid approach, combining the strengths of classical convolutional networks and quantum circuits, allows the HQSC to efficiently process and encode the input data, ultimately leading to improved performance in the sentiment classification task.

4.5 VQC as a quantum-enhanced network layer: motivation, formulation, and design

4.5.1 Motivation for quantum enhancement

Our approach is motivated by two primary hypotheses rooted in the principles of quantum machine learning:

Enhanced Expressivity: A classical neural layer typically operates on vectors $\mathbf{v}$ within an n-dimensional real vector space, $\mathbb{R}^n$. In contrast, a VQC with N qubits operates on quantum states $|\psi\rangle$ within a Hilbert space $\mathcal{H}$ whose dimension scales exponentially with the number of qubits:

$$\dim(\mathcal{H}) = 2^N \tag{9}$$

This exponential scaling means that even a modest number of qubits grants the VQC access to an exceptionally large computational space. This vast space provides the potential for VQCs to capture more complex and higher-order correlations within the data than a classical layer of a similar structural size.

Parameter Efficiency and Generalization: To map an n-dimensional input to an m-dimensional output, a classical linear layer requires a weight matrix W with $n \times m$ parameters, resulting in a complexity of $O(nm)$. A VQC performing a similarly complex transformation can be far more efficient. The number of trainable parameters in a VQC, $|\boldsymbol{\theta}|$, typically scales polynomially with the number of qubits N and the circuit depth L:

$$|\boldsymbol{\theta}| \in O(\text{poly}(N, L)) \tag{10}$$

Since the dimension of the state space (2^N) grows exponentially while the parameters grow only polynomially, a VQC can represent highly complex transformations with relatively few parameters. This efficiency can reduce the model's risk of overfitting and improve its ability to generalize to unseen data.

However, it is crucial to acknowledge that increased expressivity implies a trade-off. Research indicates that quantum circuits with excessive expressibility are prone to the "Barren Plateau" phenomenon, where gradients vanish exponentially with the number of qubits, rendering the model untrainable. Consequently, finding an optimal balance between model capability (expressivity) and trainability is a critical design constraint. In this work, we specifically address this trade-off by tailoring the circuit architectures for different layers, as detailed in the VQC Design Rationale (Sect. 4.5.3).

4.5.2 Mathematical formulation as a network layer

Each VQC in our model is formulated as an end-to-end differentiable layer that maps classical data to classical data through a quantum process. The specifics of this mapping are adapted to its architectural role, either as a fully-connected layer or a 1D convolution kernel.

When functioning as a fully-connected layer, as in our hybrid transformer module (Sect. 4.3), it maps an input vector $\mathbf{v} \in \mathbb{R}^n$ to an output vector $\mathbf{v}' \in \mathbb{R}^m$. This process begins with encoding $\mathbf{v}$ into an N-qubit initial state, $|\psi_{in}\rangle = E(\mathbf{v})$. The state is then evolved by the parameterized unitary operator of the VQC, $U_{fc}(\boldsymbol{\theta})$, according to:

$$|\psi_{\text{out}}\rangle = U_{fc}(\boldsymbol{\theta})\,|\psi_{\text{in}}\rangle \tag{11}$$

The final classical output vector $\mathbf{v}'$ is constructed by measuring the expectation values of a set of Pauli operators, where the j-th component is given by:

$$v'_j = \langle\psi_{\text{out}}|\,Z_j\,|\psi_{\text{out}}\rangle = \langle\psi_{\text{in}}|\,U^{\dagger}_{fc}(\boldsymbol{\theta})Z_j U_{fc}(\boldsymbol{\theta})\,|\psi_{\text{in}}\rangle \tag{12}$$

Alternatively, when serving as a 1D convolution kernel (Sect. 4.4), the VQC maps an input feature vector $\mathbf{x} \in \mathbb{R}^L$ to an output feature vector $\mathbf{y} \in \mathbb{R}^{L'}$. For a kernel of size k and a stride of s, the i-th element of the output vector y_i is computed by applying the quantum kernel to a patch $\mathbf{x}_i \subset \mathbf{x}$. This patch is defined as $\mathbf{x}_i = [x_{i\cdot s}, x_{i\cdot s+1}, \ldots, x_{i\cdot s+k-1}]$. The patch $\mathbf{x}_i$ is encoded into a k-qubit state $|\psi_{in,i}\rangle = E(\mathbf{x}_i)$ and processed by a shared unitary quantum kernel $U_{conv}(\boldsymbol{\theta})$. The output feature y_i is then the expectation value of a single observable, such as Z_0:

$$y_i = \left\langle\psi_{in,i}\right|\,U^{\dagger}_{conv}(\boldsymbol{\theta})Z_0 U_{conv}(\boldsymbol{\theta})\,\left|\psi_{in,i}\right\rangle \tag{13}$$

This process is repeated for all valid i to construct the full output vector $\mathbf{y}$. In both formulations, the process defines a differentiable map, allowing the VQC to be trained end-to-end with standard gradient-based optimization methods.

4.5.3 VQC design rationale

The Variational Quantum Circuits (VQCs) are the foundational building blocks of our HCQNSC model, functioning as trainable neural network layers. The specific architecture of these circuits is deliberately chosen, guided by established principles that balance

Table 1 Dataset statistics for various sentiment analysis datasets

Dataset	Number of Samples	Labels	Average Review Length
SST-2	69,170	pos/neg	19 words
IMDB	50,000	pos/neg	230 words
Yelp	6,990,280	pos/neg	180 words

a circuit's representative power against its practical trainability. This section details the theoretical rationale behind our VQC designs and provides their formal mathematical definition as a differentiable layer.

The design of our VQCs is guided by two key metrics established in the field: Expressibility and Entangling Capability. Expressibility measures a circuit's ability to generate a diverse range of quantum states, while Entangling Capability quantifies its power to create entanglement between qubits. Research indicates these two properties are often positively correlated—circuits with high expressibility tend to have strong entangling power.

However, the central challenge in VQC design lies in the trade-off between this representative power and trainability. Circuits that are too expressive are highly susceptible to the barren plateau phenomenon, where cost function gradients vanish exponentially, making optimization impossible. Our design strategy directly confronts this trade-off:

i For HQSC-I Layers: In these initial layers, which operate on a higher number of qubits, the risk of barren plateaus is significant. Therefore, we prioritize trainability by employing VQCs with simpler structures (e.g., nearest-neighbor connectivity). This choice consciously moderates the circuit's expressibility to ensure stable gradient flow, making it feasible to train a deep network.

ii For HQSC-II Layers: Conversely, in the later layers, the feature dimensionality and required qubit count are lower. With a reduced risk of barren plateaus, we can safely employ more complex circuits with richer connectivity (e.g., all-to-all). This design aims to maximize expressibility and entangling capability in a regime where the circuit remains trainable, enabling the model to capture more intricate features from the already refined data representations.

5 Experiment and result

5.1 Datasets

We conducted our experiment in three well-known binary sentiment classification benchmarks, including SST-2 (Stanford Sentiment Treebank 2), IMDB (Internet Movie Database) and Yelp Reviews. The SST-2 dataset is derived from the Stanford Sentiment Treebank, a popular dataset for sentiment analysis. It consists of movie reviews labeled as positive or negative. This binary classification setup allows us to focus on overall sentiment without additional neutral sentiments. The IMDB dataset contains movie reviews labeled with binary sentiment. It includes reviews from IMDB, each labeled as either positive or negative, making it suitable for sentiment analysis tasks requiring longer text processing. The Yelp Reviews dataset contains reviews of various businesses from Yelp, categorized into positive and negative sentiment for sentiment analysis. The dataset captures a range of opinions on service, ambiance, and experience, making it diverse and applicable to real-world sentiment classification tasks. The dataset statistic is shown in Table 1.

5.2 Evaluation metrics

To comprehensively evaluate the performance of the proposed method, Accuracy (ACC) and F1 Score (F1) are adopted as evaluation metrics.

Accuracy (ACC): Accuracy is one of the most widely used metrics for classification tasks. It measures the proportion of correctly classified instances (both positive and negative) out of the total number of instances. The formula for Accuracy is expressed as:

$$\text{ACC} = \frac{TP + TN}{TP + TN + FP + FN} \tag{14}$$

Where TP (True Positive) indicates the number of positive samples correctly predicted as positive. TN (True Negative) indicates the number of negative samples correctly predicted as negative. FP (False Positive) indicates the number of negative samples incorrectly predicted as positive. FN (False Negative) indicates the number of positive samples incorrectly predicted as negative.

F1 Score (F1): The F1 Score is a metric that combines Precision and Recall into a single value by calculating their harmonic mean. It is particularly useful when the dataset is imbalanced, as it considers both false positives and false negatives. The formula for F1 Score is given by:

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{15}$$

Where Precision indicates the proportion of true positive predictions among all positive predictions, defined as Precision = $\frac{TP}{TP+FP}$. Recall indicates the proportion of true positive predictions among all actual positive samples, defined as Recall = $\frac{TP}{TP+FN}$.

5.3 Settings

5.3.1 Comparison models

To evaluate the effectiveness of our proposed Hybrid Model HCQNSC, we selected comparison models from three main categories:

1. Non-transformer-based Neural Network Models: These models represent traditional deep learning architectures for NLP tasks and include:
   - TextCNN [37]: A convolutional neural network model tailored for text classification tasks (Kim, 2014). We implemented the CNN-non-static variant, which initializes the word vector matrix using pre-trained embeddings, achieving a balance between computational efficiency and contextual representation.
   - GRU [38]: Gate Recurrent Unit (Chung et al., 2014), a streamlined recurrent neural network that reduces computational overhead and accelerates training while capturing sequential dependencies, making it advantageous over standard RNNs.
2. Transformer-based Models: To compare against transformer architectures directly analogous to our HCQNSC design, we developed a Classical Transformer-based Neural Network for Sentiment Classification (CTNSC). The CTNSC replicates the structural configuration of HCQNSC, with the only modification being the substitution of quantum-specific components (HQSC-I and HQSC-II layers) with classical transformer layers. This allows for a direct comparison between the quantum-enhanced HCQNSC and its purely classical counterpart.

Table 2 Model parameter settings

	HCQNSC	CTNSC	Qtransformer	QTC	GRU	TextCNN
Max epochs	10	10	10	10	10	10
Batch size	8	32	8	8	32	32
Embedding dim	1024	1024	8	1024	1024	512
Learning rate	8e-6	1e-3	1e-5	1e-5	1e-3	1e-3
Dropout	0.2	0.2	0.2	0.2	0.2	0.2
No. of layers	5	5	5	5	5	5

3. Classical-Quantum Hybrid Classification Models: We included existing classical-quantum hybrid models to benchmark HCQNSC against similar architectures that incorporate quantum enhancements in NLP:
    - QTC [29]: A hybrid classical-quantum model consists of a novel random quantum temporal convolution (QTC) learning framework replacing some layers in the BERT-based decoder.
    - Qtransformer [28]: A quantum-enhanced transformer-based model that replaces the linear transformation with VQCs.

These comparison models provide a comprehensive baseline for assessing the performance of HCQNSC in sentiment classification tasks, offering insights into the added value of quantum-enhanced layers relative to both traditional and hybrid quantum-classical NLP models.

5.3.2 Experiment details

The implementation of our proposed hybrid model is grounded on TensorFlow and PennyLane, with TensorFlow responsible for the classical components and PennyLane managing the quantum operations. For our experiment setup, we utilized Amazon Braket Hybrid Jobs on AWS, enabling the integration of GPU and Quantum simulator resources provided by Amazon. This setup facilitated the efficient deployment and scaling of our hybrid model, leveraging quantum capabilities where applicable within the overall architecture.

The quantum circuit configurations employed in our model featured qubit counts ranging from 3 to 10. Hyperparameters for each model included in the comparative study are outlined in Table 2. During the training phase, cross-entropy loss was used as the loss function, optimized with the Adam optimizer for parameter tuning. We implemented a grid search to fine-tune key parameters such as batch size $(8, 16, 32, 64)$, learning rate $(10^{-4}, 10^{-5}, 10^{-6})$, and dropout rates $(0.1, 0.15, 0.2, 0.25)$ to determine the best configurations for model convergence and performance consistency. The final model configurations were selected based on performance benchmarks on the validation dataset.

For the evaluation of our hybrid model and the comparison models, we employed accuracy and F1-score as primary metrics. These metrics were chosen to provide a balanced assessment of model performance, accounting for both the correctness of predictions and the model's ability to maintain performance consistency across positive and negative sentiment classifications.

5.4 Results

We conducted a thorough comparison between the proposed HCQNSC model and various models, including traditional neural network models and other classical-quantum

Table 3 Performance comparison of different methods

Method	SST-2		IMDB		Yelp	
	Acc(%)	F1(%)	Acc(%)	F1(%)	Acc(%)	F1(%)
TextCNN	82.23	81.83	86.39	86.15	87.13	87.07
GRU	83.47	83.61	87.33	87.24	88.74	88.69
CTNSC	92.26	92.18	**93.46**	**93.42**	92.76	92.65
Qtransformer	88.32	88.24	89.10	88.93	89.69	89.47
QTC	89.18	89.12	89.93	89.67	90.32	89.84
HCQNSC	**92.88**	**92.76**	93.28	92.97	**92.89**	**92.84**

hybrid models, to assess their efficacy in sentiment classification tasks across the SST-2, IMDB, and Yelp datasets. Table 3 reports the best performance achieved across all runs.

Traditional Neural Network Models (Non-Transformer-based): The non-transformer neural network models evaluated are TextCNN, a convolutional neural network optimized for text classification, and GRU, a gated recurrent unit model tailored for sequence dependencies.

Both models performed in the lowest tier as expected, with accuracy and F1 scores differing from the second-tier models by more than 6% at the maximum and over 1% at the minimum.

Compared to the top-tier models, their accuracy and F1 scores showed a gap of over 10% at the maximum and no less than 4% at the minimum. While these models capture some level of contextual relationships, they lack the depth and complexity found in transformer-based architectures, limiting their accuracy and F1 scores on sentiment classification tasks.

Transformer-based Models: The transformer-based models in our comparison include our proposed HCQNSC and its purely classical counterpart, the Classical Transformer-based Neural Network for Sentiment Classification (CTNSC). The CTNSC is designed with an equivalent 5-layer, dimensionality-reducing structure for a direct and fair comparison. As shown in Table 3, which reports the best performance achieved, both models deliver top-tier results with only marginal differences between them. To rigorously evaluate this comparison, we analyzed both the parameter efficiency of the core transformer layers and the statistical significance of the performance results.

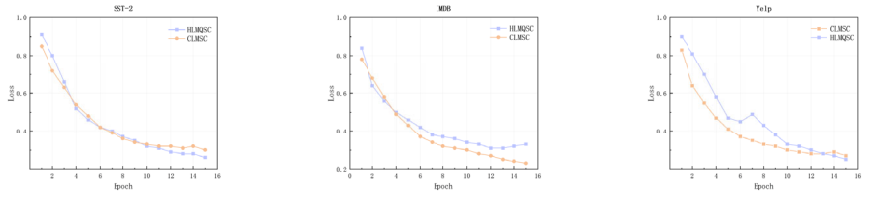
Our primary claim is that the HCQNSC architecture offers a dramatic advantage in parameter efficiency. Theoretically, by replacing classical linear layers of complexity $O(n^2)$ with our VQC design—and leveraging Amplitude Encoding for high-dimensional inputs—the parameter complexity for these core transformations can be reduced to $O(\text{poly}(\log n))$. This theoretical advantage is reflected in the actual model implementations. Depending on the hyperparameter settings, the total number of trainable parameters for the core components (Q, K, V, O, and FFN layers) in our HCQNSC model ranges from approximately 600 to 1500. In stark contrast, the parameter count for the same components in the classical CTNSC model ranges from 0.5 million to 1.28 million.

Given this vast difference in parameter complexity, we then performed a statistical analysis based on 10 independent runs to determine if the performance was comparable. In Table 4, the results show a nuanced picture:

- On the SST-2 dataset, our HCQNSC model's performance is statistically superior to that of the CTNSC (for both Acc and F1, $p < 0.05$).

Table 4 Statistical Performance Comparison of HCQNSC and CTNSC models. The comparison is based on 10 independent runs with paired random seeds. The p-value is from a two-tailed paired t-test. **Bold** values indicate a statistically superior result ($p < 0.05$)

Dataset	Metric	HCQNSC (mean ± std)	CTNSC (mean ± std)	p-value	Significant?
SST-2	Acc (%)	**92.75 ± 0.08**	92.16 ± 0.07	< 0.05	Yes
	F1 (%)	**92.68 ± 0.06**	92.10 ± 0.05	< 0.05	Yes
IMDB	Acc (%)	93.12 ± 0.08	**93.37 ± 0.05**	< 0.05	Yes
	F1 (%)	92.88 ± 0.06	**93.36 ± 0.04**	< 0.05	Yes
Yelp	Acc (%)	92.79 ± 0.06	92.73 ± 0.05	> 0.05	No
	F1 (%)	92.75 ± 0.07	92.67 ± 0.06	> 0.05	No



(a) Training Loss Curve on SST-2 Dataset (b) Training Loss Curve on IMDB Dataset (c) Training Loss Curve on Yelp Dataset

Figure 6 Comparison of Training Loss Curves for HCQNSC and CTNSC models

- On the IMDB dataset, the classical CTNSC's performance is statistically superior (for both Acc and F1, $p < 0.05$).
- On the Yelp dataset, there is no statistically significant difference between the two models' performance (for both Acc and F1, $p > 0.05$).

Therefore, these combined analyses confirm our central finding: the proposed HCQNSC architecture achieves an exponential advantage in parameter efficiency. In exchange for this, its performance, when compared to a classical counterpart, is highly competitive and dataset-dependent—ranging from statistically superior to comparable, and in some cases, slightly inferior. This positions HCQNSC as a powerful and resource-efficient alternative for sentiment classification tasks.

Classical-Quantum Hybrid Models: The comparison also included hybrid models like QTC, which integrates quantum convolutional layers, and Qtransformer, which replaces the linear transformation parts of a transformer with quantum circuits.

These models showed moderate performance, with results between those of the non-transformer neural network models and the transformer-based CTNSC and HCQNSC. Although QTC and Qtransformer did not exceed the performance of the classical transformer-based CTNSC, they achieved similar results (with the maximum gap in accuracy and F1 scores being within 4.5%) with a significantly reduced parameter count, underscoring the efficiency of quantum circuits. Their performance, while not superior, suggests that quantum models can offer competitive outcomes with lower computational requirements, positioning them as viable alternatives under resource constraints.

Comparison of Training Loss Curves for HCQNSC and CTNSC models among three datasets are shown in Fig. 6. Overall, the HCQNSC and CTNSC achieved the best results among the tested models, confirming the effectiveness of transformer-based architectures

Table 5 Comparison of HCQNSC Performance under Noiseless and Noisy Simulation on the SST-2 Dataset

Model Type	Training Failure Rate	Best Accuracy (%)	Best F1-Score (%)
Noiseless HCQNSC	0% (0/10 runs)	92.88	92.76
Noisy HCQNSC	50% (5/10 runs)	84.52	84.17

for sentiment classification. The incorporation of quantum circuits in HCQNSC demonstrated a notable, if not drastic, improvement, showing potential for future development as quantum hardware capabilities advance.

5.5 Performance analysis under noise conditions

We conducted a comparative analysis under both ideal (noiseless) and realistic noise conditions. This analysis aims to quantify the impact of noise not only on the model's optimal performance but also on the stability and success of the training process.

For this experiment, we created a Noisy HCQNSC model using the TensorCircuit library. We introduced a depolarizing noise channel after each gate, with error probabilities (p) set based on the calibration data of a real IBM Quantum device. We then compared this noisy model against the ideal, Noiseless HCQNSC on the SST-2 dataset over 10 independent runs. A training run is defined as a "failure" if its final validation accuracy does not significantly improve over random guessing (i.e., remains below 50%), a common indicator of untrainable models stuck in barren plateaus.

The results of this comparative experiment are summarized in Table 5.

As shown in Table 5, the introduction of noise has two significant effects. First, it critically impacts the reliability of the training process. The ideal noiseless model was trainable in all runs, exhibiting a 0% failure rate. However, for the noisy model, 5 out of 10 runs failed to meet the success criterion. An analysis of the training curves for these failed runs reveals a consistent pattern: the loss value would stagnate after a few initial epochs and remain effectively unchanged. This behavior is a classic symptom of vanishing gradients, providing direct evidence that hardware noise exacerbates the barren plateau phenomenon in our VQCs.

Second, noise dramatically degrades the model's optimal performance. Among all successful runs, the best accuracy achieved by the noisy model plummeted to 84.52%, a stark contrast to the 92.88% achieved by the noiseless model. This substantial performance degradation quantifies the severe impact of hardware noise on the model's predictive capability. These findings underscore that overcoming noise-induced training failures and performance loss is a key challenge for the practical application of such architectures.

5.6 Analysis of model complexity and performance

In this section, we experimentally investigate how different aspects of our model's complexity affect its final performance. We analyze three key factors: the initial number of qubits which defines the model's scale, the inclusion of the hybrid convolutional module as a core component, and the distribution of different layer types which governs the balance between the model's expressive power and its trainability.

5.6.1 Effect of different initial numbers of qubits

In quantum neural networks, the number of qubits significantly influences model complexity and susceptibility to the barren plateau phenomenon. However, in our proposed

Table 6 Comparison of Num of Qubits

# of Qubits	SST-2		IMDB		Yelp	
	Acc(%)	F1(%)	Acc(%)	F1(%)	Acc(%)	F1(%)
8	91.52	91.23	91.79	91.42	91.76	91.68
9	90.82	90.58	90.92	90.66	90.80	90.72
10	**92.88**	**92.76**	**93.28**	**92.97**	**92.89**	**92.84**

model, the number of qubits within the quantum circuits is not fixed throughout the network but decreases progressively layer by layer. As a result, we specifically evaluated the impact of the initial number of qubits in the first layer of the quantum circuit on overall model performance.

We maintained the total number of layers in the model as five, with a distribution of two HQSC-I layers and three HQSC-II layers—a configuration verified in the subsequent section to be optimal. The experiments were conducted with initial qubit counts of 10, 9, and 8 in the first layer, while keeping the qubit counts in subsequent layers consistent across all configurations. The model performance was then assessed on three datasets: SST-2, IMDB, and Yelp.

The results are summarized in Table 6. Taking the model's final choice of 10 initial qubits as a baseline, reducing the initial qubit count to 9 led to a significant performance drop across all three datasets, with decreases ranging between 2% and 2.4%. Further reducing the initial qubit count to 8 resulted in a performance rebound, with improvements ranging between 0.65% and 1% compared to the 9-qubit configuration. However, the performance with 8 qubits still fell short of the 10-qubit baseline, showing gaps of 1.1% to 1.6%. These results suggest that Higher Initial Qubits (10) may provide greater expressivity and entangling capability, enabling the model to better capture and process complex linguistic structures. Lower Initial Qubits (8), though slightly less expressive, may show competitive performance due to its reduced susceptibility to barren plateaus in deeper layers. Intermediate Initial Qubits (9) may suffer from a suboptimal balance between expressivity and trainability, leading to its comparatively lower performance. These findings highlight the importance of carefully choosing the initial qubit count to balance the trade-offs between expressivity and optimization challenges.

5.6.2 Effect of the hybrid convolutional module

The hybrid convolutional module was introduced into the HCQNSC architecture with the goal of reducing the model's overall complexity, thereby enhancing its trainability and final performance. To experimentally verify whether this module achieves its intended effect of improving the training outcome, we conducted an ablation study. We created an "ablated" version of our best-performing HCQNSC model, which is identical in every aspect except for the complete removal of the hybrid convolutional module from the HQSC-II layers. We then compared the performance of the full model against this ablated version on the SST-2 dataset.

The results are presented in Table 7. As the table shows, removing the hybrid convolutional module leads to a consistent and significant performance drop across all benchmarks. The performance degradation is most pronounced on the IMDB dataset, with a drop in accuracy of over 3.5%. For the SST-2 and Yelp datasets, the accuracy decreases by approximately 2.5% and 2.2% respectively. A similar degradation is observed for the

Table 7 Ablation Study of the Hybrid Convolutional Module Across All Datasets

Model Version	SST-2		IMDB		Yelp	
	Acc (%)	F1 (%)	Acc (%)	F1 (%)	Acc (%)	F1 (%)
Full HCQNSC	**92.88**	**92.76**	**93.28**	**92.97**	**92.89**	**92.84**
HCQNSC (w/o Conv. Module)	90.35	90.21	89.75	89.58	90.65	90.58

Table 8 Performance with Varying Layer Distributions on the SST-2 and IMDB Datasets

Configuration	SST-2		IMDB	
	Acc (%)	F1 (%)	Acc (%)	F1 (%)
α-1 (1:1)	89.55	89.41	90.15	90.02
α-2 (1:2)	91.30	91.15	91.80	91.66
α-3 (1:3)	92.10	92.02	92.70	92.61
α-4 (1:4)	**92.33**	**92.25**	**92.97**	**92.86**
α-5 (1:5)	90.55	90.41	90.80	90.65
β-1 (2:1)	91.20	91.08	91.95	91.82
β-2 (2:2)	92.15	92.03	92.75	92.63
β-3 (2:3)	**92.88**	**92.76**	**93.28**	**92.97**
β-4 (2:4)	91.25	91.13	91.50	91.38

F1-scores. This ablation study shows that the hybrid convolutional module makes a substantial and positive contribution to the model's performance, making it an effective mechanism for feature compression and refinement.

5.6.3 Effect of different layer distributions

To provide a theoretical explanation for the optimal architectural depth and complexity, this section investigates the trade-off between the model's expressivity and its trainability. We achieve this by analyzing how performance varies with the layer distribution. In this context, "layer distribution" has a dual meaning: it refers not only to the total depth of the network but also to the proportion of HQSC-II layers—which contain the complexity-reducing hybrid convolutional module—within the overall architecture.

We designed two series of experiments, conducted on both the SST-2 and IMDB datasets. In the first series (α-series), we fixed the number of HQSC-I layers at one and varied the number of HQSC-II layers from one to five. In the second series (β-series), we fixed the HQSC-I layers at two and varied the HQSC-II layers from one to four. The best-run performance for each configuration is reported in Table 8.

The results in Table 8 illustrate the expressivity-trainability trade-off, showing a consistent trend across both datasets. In both the α and β series, performance initially improves as more HQSC-II layers are added, peaking at an optimal point for each series (α-4 and β-3). This is because a deeper, more complex network possesses higher expressivity, allowing it to capture more intricate features from the data.

However, after reaching this optimal point, adding more complex layers leads to a sharp decline in performance. As seen in configurations α-5 and β-4, the accuracy drops significantly. This degradation occurs because the overly expressive circuits become highly susceptible to the barren plateau phenomenon, which diminishes trainability due to vanishing gradients, leading to suboptimal convergence.

This experiment shows that the optimal architecture for a hybrid quantum model is not simply the deepest or most complex one, but rather one that strikes a delicate balance

between being expressive enough to solve the task and simple enough to remain trainable. Our best-performing 2:3 layer configuration (β-3) precisely represents this optimal balance.

6 Conclusion

In this study, we introduced a Hybrid Neural Network (HCQNSC) for sentiment classification that incorporates VQCs and quantum convolutional layers within a transformer-based framework. In our experiments, the model outperformed the evaluated non-transformer-based classical neural networks in terms of both accuracy and F1 score across multiple datasets, demonstrating its ability to capture sentiment features more effectively.

While HCQNSC shows competitive performance compared to classical transformer models, it does not universally surpass them across all datasets. Nonetheless, it achieves results comparable to or slightly below classical transformers with a significantly lower parameter count due to the integration of quantum circuits. This efficiency is made possible by the unique properties of quantum computing, enabling complex transformations within a compact representation. Additionally, by designing quantum circuits with specific structures for different layers and incorporating a quantum convolutional module, HCQNSC surpasses existing hybrid models in performance and efficiency, highlighting the benefits of hybrid quantum-classical architectures in NLP.

Furthermore, our analysis of the model's complexity revealed critical insights for designing hybrid architectures. The experiments on varying the initial number of qubits and the distribution of HQSC layers, presented in Sect. 5.6, did more than just identify an optimal hyperparameter configuration. Specifically, the analysis of layer distributions provided strong empirical evidence for the expressivity-trainability trade-off, demonstrating how performance degrades when a model becomes too complex and susceptible to the barren plateau phenomenon. Understanding this balance is crucial for developing practical quantum-enhanced models.

Looking forward, we plan to explore more VQC architectures designed explicitly around this principle. A key limitation of this study, however, is that while we have performed noise-aware simulations (Sect. 5.5) to assess the model's robustness, these experiments were ultimately conducted on classical simulators. Therefore, a crucial next step for future work is to deploy and validate the HCQNSC model on genuine quantum hardware, which will involve tackling challenges like noise mitigation and circuit compilation. Through these future explorations, we aim to further establish the practical utility of hybrid quantum-classical architectures for NLP tasks in the NISQ era.

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Author contributions
Tingyu Liu, as the first author, contributed to the proposal of the method, its implementation, experimental validation, and the writing of the original manuscript. Ying Li was responsible for the revision of the manuscript. Jingtao Wang and Yun Pan, as corresponding authors, supervised the work.

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Data availability
The datasets used in this study, including IMDB, Yelp, and SST-2, are publicly available. The links to access these datasets are as follows: IMDB dataset: https://ai.stanford.edu/amaas/data/sentiment/, Yelp dataset: https://www.yelp.com/dataset, SST-2 dataset: https://nlp.stanford.edu/sentiment/. These datasets can be freely accessed for research purposes.

Declarations

Competing interests
The authors declare no competing interests.

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